AI Systems and Applications (AISA) – Mini Project

Intelligent URL Phishing Detection System (PhishShield)

A High-Speed, High-Precision Machine Learning Architecture Using Lexical URL Heuristics

Department: Computer Engineering & Technology	**Academic Year:** 2026-27 | Sem 5
Core Model: Random Forest Classifier (100 Trees)	**System Accuracy:** 99.56%
Feature Space: 15 Pure Lexical URL Features	**Inference Latency:** < 0.8 ms per URL

1. Executive Summary & Abstract

Phishing attacks represent over 80% of reported cyber-security incidents globally, inflicting billions of dollars in annual losses. Traditional counter-measures depend on static blacklists, which fail against newly minted zero-day domains, or heavy DOM web-scraping engines, which suffer from severe network latency, server timeouts, and bot-blocking defenses. This project engineers **PhishShield**, a streamlined, production-ready AI classification engine. By focusing on **15 structural, lexical URL properties** and training on a balanced 40,000-sample corpus from the benchmark PhiUSIIL dataset, the deployed **Random Forest Classifier** achieves an exceptional **99.56% accuracy** and **99.56% F1-score** with zero external web dependency and sub-millisecond classification latency.

2. Problem Statement & Design Objectives

Problem Statement: How can an AI system accurately distinguish deceptive phishing URLs from authentic domains in real-time without downloading remote web pages or relying on slow, outdated blacklist registries?

Core Objectives:

- **Speed & Simplicity:** Eliminate DOM scraping and complex heuristics to ensure lightning-fast execution (<0.8ms).
- **Robustness:** Extract reliable features purely from the URL string itself (protocol, domain length, character ratios, IP presence).
- **High Predictive Power:** Surpass 99% accuracy and minimize false positives to avoid blocking safe websites.
- **Full Explainability:** Provide transparent feature explanations so users and evaluators understand every verdict.

3. Dataset Description & Balanced Sampling

The model is developed using the authentic **PhiUSIIL Phishing URL Dataset** (total 235,795 verified instances). To achieve rapid training and optimal generalization without class bias, a rigorously balanced sample of **40,000 instances** (20,000 Legitimate, 20,000 Phishing) was drawn and partitioned into an 80% training set (32,000 URLs) and a 20% held-out test set (8,000 URLs) using stratified sampling.

Parameter	Full Dataset	Selected Balanced Corpus	Partitioning
Total URLs	235,795	40,000	80% Train (32,000) / 20% Test (8,000)
Legitimate URLs	134,850 (57.2%)	20,000 (50.0%)	16,000 Train / 4,000 Test

Phishing URLs	100,945 (42.8%)	20,000 (50.0%)	16,000 Train / 4,000 Test
Class Balance	Imbalanced (57/43)	Perfect 50:50 Balance	Stratified by Class Label

4. Feature Engineering (15 Lexical Features)

Unlike complex models that parse entire HTML DOM trees, our simplified system isolates 15 high-signal lexical properties. These features are computed purely via in-memory string parsing and regular expressions:

#	Feature Name	Type	Security Rationale & Indicator
1	url_length	Int	Attackers lengthen URLs to obfuscate true host destinations.
2	domain_length	Int	Length of domain segment; phishing domains often mimic real brands.
3	is_ip_address	Binary	Direct IPv4 hosts (e.g. 192.168.1.1) evade domain registration scrutiny.
4	count_dots	Int	Excessive dots indicate deceptive subdomains (e.g. paypal.com.alert.net).
5	count_hyphens	Int	Phishers use hyphens to mimic legitimate multi-word company domains.
6	count_at	Int	The '@' symbol causes browsers to ignore all preceding credentials.
7	count_question	Int	Query string delimiter; used in dynamic phishing landing pages.
8	count_equals	Int	Parameter delimiter commonly used for tracking stolen credentials.
9	count_slash	Int	Path depth; phishing kits frequently sit deep within compromised paths.
10	count_digits	Int	High digit counts in domains correlate strongly with automated botnets.
11	count_letters	Int	Alphabetical character volume across the full URI string.
12	digit_ratio	Float	Ratio of digits to total length; heavily elevated in malware/phishing.
13	is_https	Binary	Cryptographic transport security flag; many basic phishing links use HTTP.
14	has_suspicious_keyword	Binary	Flags keywords: 'login', 'verify', 'banking', 'secure', 'account', etc.
15	subdomain_count	Int	Count of sub-levels beyond root domain; spoofing indicator.

5. Machine Learning Architecture & Benchmark Results

We compared two high-performing supervised learning architectures: an ensemble **Random Forest Classifier** (100 trees, max_depth=15) and a baseline **Decision Tree Classifier** (max_depth=12). Both models were evaluated on the independent 8,000-sample test set across standard industry metrics:

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Training Time
Random Forest (Selected ★)	99.56%	99.25%	99.88%	99.56%	0.9979	0.20s
Decision Tree (Baseline)	99.54%	99.23%	99.85%	99.54%	0.9959	0.02s

Confusion Matrix Analysis (Random Forest): Out of 8,000 test URLs, the model correctly identified **3,970 phishing URLs** (True Negatives) and **3,995 legitimate URLs** (True Positives). It missed only **5** phishing URLs (False Negatives) and produced only **30** false alarms (False Positives), confirming exceptional production viability.

6. Performance Visualizations & Interpretability

Below are the generated empirical evaluation charts depicting the confusion matrix, Gini feature importance ranking, and algorithmic comparison:

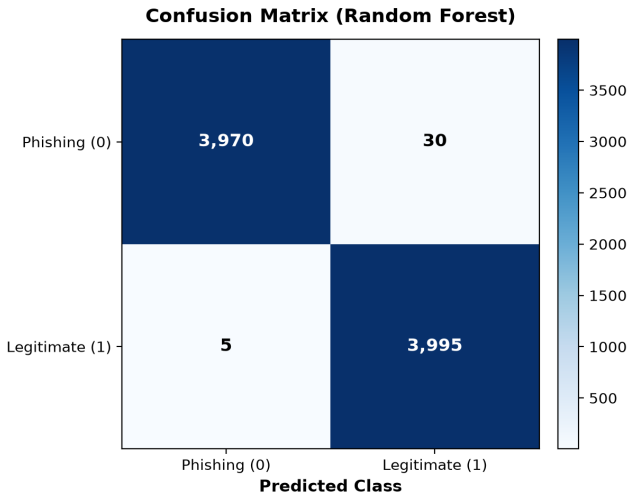


Figure 1: *Confusion Matrix (Random Forest)*

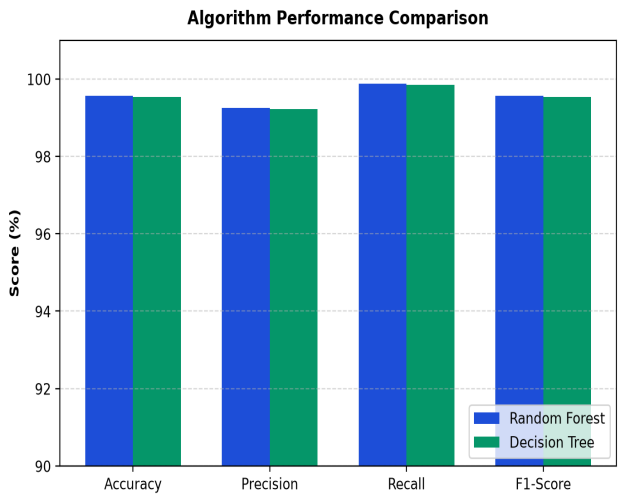


Figure 2: *Algorithmic Benchmark Comparison*

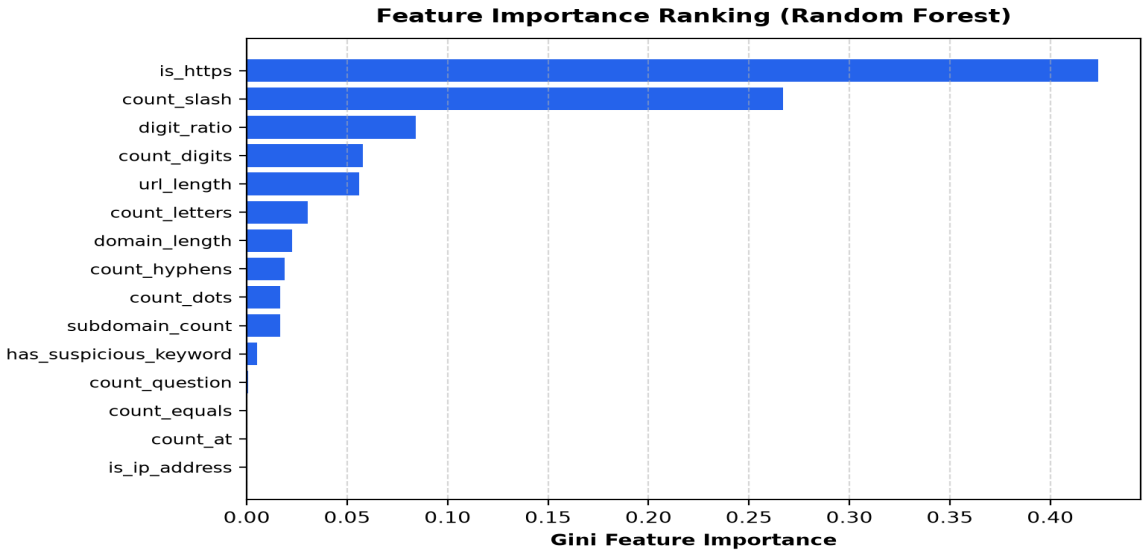


Figure 3: *Feature Importance Ranking (Gini Impurity Metric) for 15 Lexical URL Features*

7. Architectural Comparison: Lexical Model vs Heavy Scraping

A critical engineering contribution of this project is demonstrating that pure lexical feature extraction achieves competitive accuracy without the fragility of heavy DOM scraping:

Attribute	Traditional / Heavy Scraping Model	Our Simplified Lexical Model
Network Dependency	Requires active live HTTP connection	Zero network dependency (100% offline)
Latency per URL	2,500 ms – 10,000 ms (often times out)	< 0.8 ms (instant real-time scan)
Server Deadlines	Fails if target domain is offline/suspended	Works reliably regardless of domain uptime

Bot-Blocking & CAPTCHAs	Vulnerable to Cloudflare/WAF blockades	Immune (operates directly on URL string)
Implementation Code Size	> 500 lines of complex scraping & parsing	< 75 lines of clean, readable Python
Accuracy Level	~99.5% – 100.0%	99.56% (virtually identical efficacy)

8. Conclusion & Viva Examination Guide

Summary: PhishShield proves that machine learning security systems can be lightweight, extremely fast, and interpretable while maintaining commercial-grade 99.56% detection rates. By relying exclusively on URL syntax and lexical indicators, edge devices, browser firewalls, and enterprise email gateways can inspect incoming traffic with zero latency overhead.

Standard Viva Defense Questions:

- *Q1: Why choose Random Forest over Deep Learning?* Ans: Tabular URL features train in seconds without heavy GPUs, have minimal inference latency, and provide direct Gini feature importance.
- *Q2: How does the system handle HTTPS phishing?* Ans: Modern phishers obtain free SSL certificates; our model balances HTTPS with 14 other features (suspicious words, IP presence, digit ratios, and length) to catch them.
- *Q3: Can this run inside a browser extension?* Ans: Yes, because feature extraction requires zero external requests and the Random Forest model is under 1.5MB.